

Total No. of Questions : 6]

SEAT No. :

P5117

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B.E./Insem.- 524

B.E. (Mechanical Engg.) (Semester - I)

ENERGY AUDIT AND MANAGEMENT (Elective - I)
(2012 Pattern)

Time :1 hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Figures to the right indicate full marks.*
- 2) *Draw neat figures wherever necessary.*
- 3) *Use of Scientific Calculators is allowed.*

- Q1)** a) Discuss Indian energy scenario in context of natural gas, oil and coal. [5]
b) List down the responsibilities of Energy Manager. [5]

OR

- Q2)** a) Name the designated consumers under the energy conservation act. [4]
b) Write a note: Walk through energy audit. [6]

- Q3)** a) Explain Energy Audit Methodology for Ice factory. [6]
b) Write short note on the following (i) Key features of Electricity Act 2003
(ii) System Distribution losses. [4]

OR

- Q4)** a) What are the energy conservation opportunities in Refrigeration and HVAC systems. [6]
b) List the different instruments used for energy audit along with their applications. [4]

- Q5)** a) A sum of Rs. 4,00,000 is deposited in a bank at the beginning of a year. The bank pays 5% interest annually. How much money is in the bank account at the end of the seventh year, if no money is withdrawn? [5]
b) What is the NPV of a project, (life 2 year) which requires an investment of Rs. 70000 and yield Rs. 20000 in the 1st year and Rs. 50000/- in the next year, if the Interest rate is 10%. [5]

P.T.O.

OR

- Q6)** Evaluate the financial merit of a proposed project shown in table below. Consider annual discount rate of 8% for each project. Use Net present analysis technique. **[10]**

	Project - I	Project - II
Capital cost (Rs.)	40,000	40,000
Year	Net Annual saving (Rs.)	Net Annual saving (Rs.)
1	6000	6600
2	6000	6600
3	6000	6300
4	6000	6300
5	6000	6000
6	6000	6000
7	6000	5700
8	6000	5700
9	6000	5400
10	6000	5400
11	6000	5000

